

Activity- 6

Case 1

AIM:

To study the nature and size of the image formed by a convex lens on a screen by using a candle and a screen (for different distances of the candle from the lens)

APPARATUS

Convex lens, candle, lens holder, burning candle, cardboard screen

THEORY:

According to the lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Where f is the focal length of the lens, u is the object distance, and v is the image distance

PROCEDURE:

1. Find the rough focal length of the convex lens by using the focusing method
2. Mount the convex lens in the lens holder. Mount the cardboard screen on another upright and keep it at a distance equal to the rough focal length of the lens from the lens.
3. Adjust heights so that the inverted image of the erect flame of a burning candle is formed on the screen. Move the screen to make the image sharp. The screen will be nearly at the focus of the convex lens.
4. As the burning candle is moved towards the lens on one side, the screen must be moved away from the lens on the other side for getting sharp flame image. The inverted image size increases.
5. Record all the observations in tabular form

RESULT:

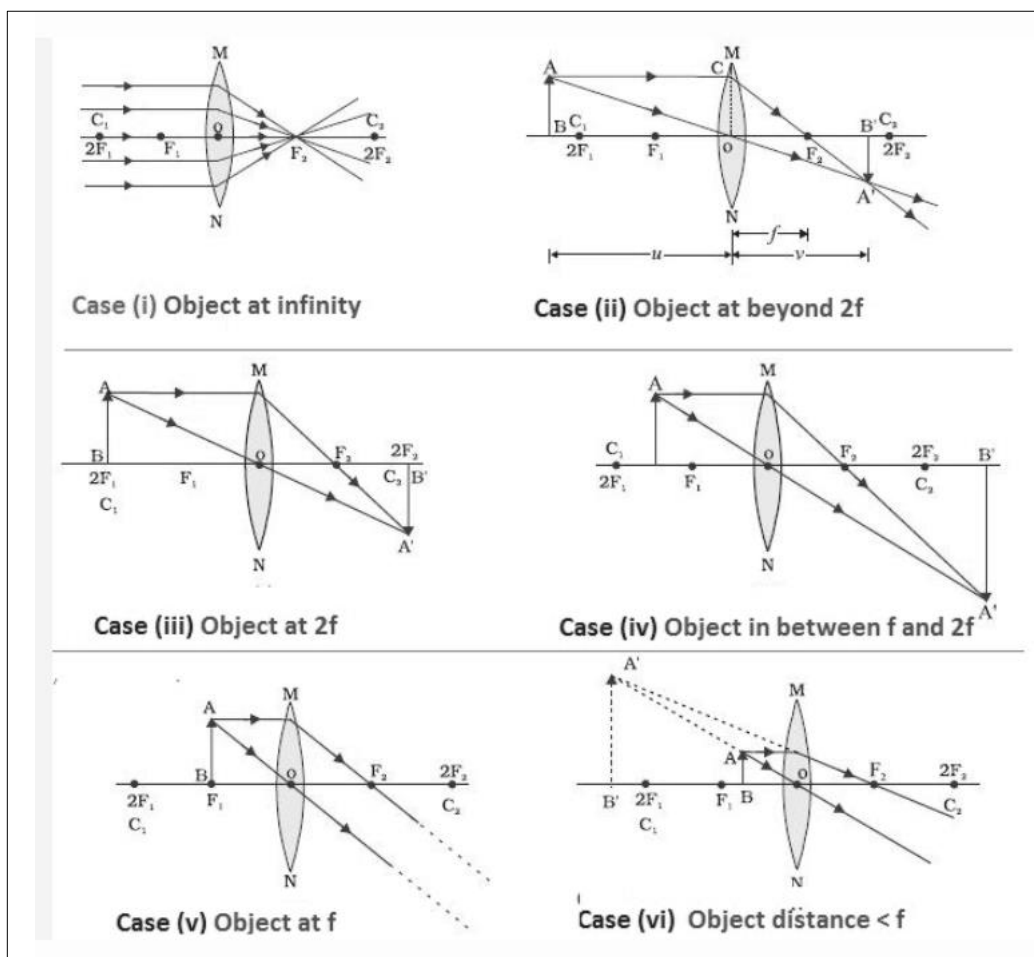
As the position of the lighted candle is changed, the change in position, nature, and size of the image is according to theoretical predictions.

Precautions and sources of error

1. The upright holding objects and images should be aligned properly.
 2. Hold the candle very carefully. Otherwise, the candle may burn you.
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Diagram and observation table on the left side of the page

Ray Diagrams:



Observation :

Focal length of the lens using the distant object method=

Position of object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus	Highly diminished	Real and inverted
Beyond 2F	Between F and 2F(C)	Point size diminished	Real and inverted
At 2F	At 2F	Same size	Real and inverted
Between F and 2F	Beyond 2F	Magnified	Real and inverted
At F	At infinity	Highly magnified	Real and inverted
Between F and pole	On the same side of the lens as object	Magnified	Virtual and erect

Write case 2 on a fresh page (new page)

Case 2:

Aim:

To study the nature and size of the image formed by a concave mirror on a screen by using a candle and screen (for different distances of the candle from the mirror)

APPARATUS

Concave mirror, candle, mirror holder, burning candle, cardboard screen

THEORY:

According to the mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

Where f is the focal length of the lens, u is the object distance, and v is the image distance

PROCEDURE:

1. Find the rough focal length of the concave mirror by using the focusing method
2. Mount the concave mirror in the mirror holder. Mount the cardboard screen on another upright and keep it at a distance equal to the rough focal length of the mirror from the mirror.
3. Adjust heights so that the inverted image of the erect flame of a burning candle is formed on the screen. Move the screen to make the image sharp. The screen will be nearly at the focus of the concave mirror.
4. As the burning candle is moved towards the mirror on one side, the screen must be moved away from the mirror on the same side for getting sharp flame image. The inverted image size increases.
5. Record all the observations in tabular form

RESULT:

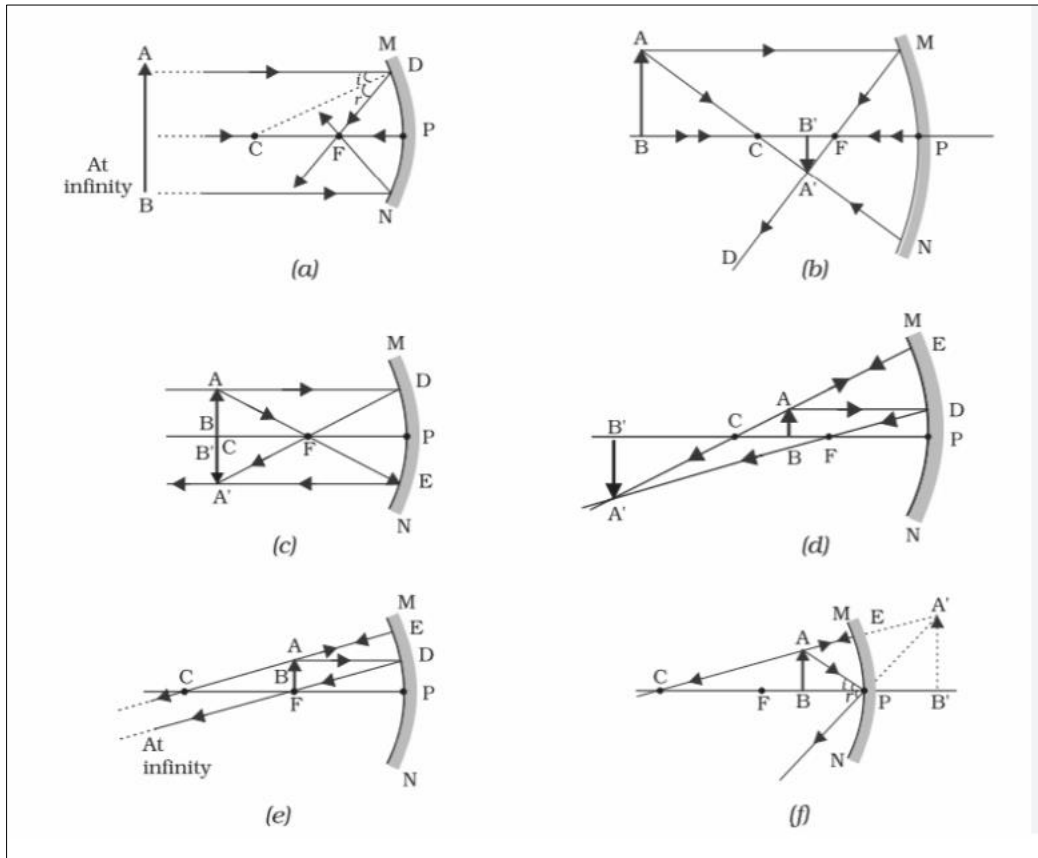
As the position of the lighted candle is changed, the change in position, nature, and size of the image is according to theoretical predictions.

Precautions and sources of error

6. The upright holding objects and images should be aligned properly.
7. Hold the candle very carefully. Otherwise, the candle may burn you.

Diagram and observation table on the left side of the page

Ray diagrams



Observation :

Focal length of the mirror using the distant object method=

Position of object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus	Highly diminished	Real and inverted
Beyond C	Between C and F	Diminished	Real and inverted
At C	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Highly enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect